KU LEUVEN

FACULTEIT INGENIEURSWETENSCHAPPEN



Sync vs async execution

Comparison

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Academiejaar 2023-2024

** There are no code examples in this text, please look in github repository under session5: github link **

Introduction:
This exercise consists of 2 parts where we see what impact async execution has on execution time. The excercise consists of making 4 arrays, filling them up with random values, and using the reduction kernel from session 2 to calculate the sum, product, min and max of the respective arrays. Both implementations do 1k iterations and only over the last 100 is the mean execution time calculated.

Part 1 (code.cu)
For the first part of the exercise we start to time at the start of the program. All memory is allocated and arays are initialised one by one. Then the 4 kernels are called one after another (after cudaDeviceSynchronize() to make everything synchronous) Finally the mean execution time is printed.

For 4 elements the mean execution time is 0,83ms, and for 10k elements it is 1,37ms.

Part 2 (code_part_2.cu)
This time memory copying and array generation happens async. While the GPU is executing a kernel, the cpu spends it's time generating the array for the next kernel. This leads to overall faster execution times.

For 4 elements the mean execution time is 0,55ms, and for 10k elements it is 1,28ms.

